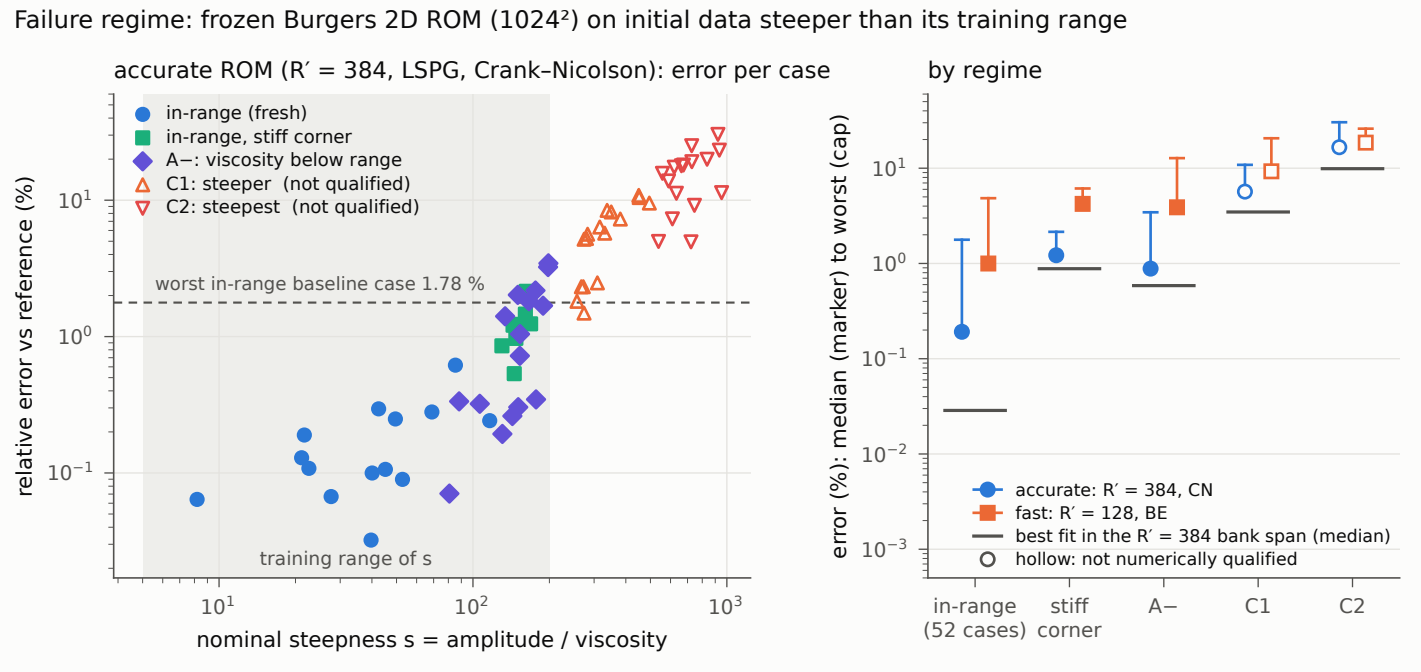
Failure regime: frozen Burgers 2D ROM (1024²) on initial data steeper than its training range
accurate ROM (R′ = 384, LSPG, Crank–Nicolson): error per case
in-range (fresh)
in-range, stiff corner
A−: viscosity below range
C1: steeper (not qualified)
C2: steepest (not qualified)
worst in-range baseline case 1.78 %
training range of s
relative error vs reference (%)
nominal steepness s = amplitude / viscosity
by regime
error (%): median (marker) to worst (cap)
accurate: R′ = 384, CN
fast: R′ = 128, BE
best fit in the R′ = 384 bank span (median)
hollow: not numerically qualified
in-range (52 cases)
stiff corner
A−
C1
C2